PowerModels Accelerated Mitigation Plan

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Approach: Agent-assisted parallel execution with expert team

Thesis

The full mitigation plan (Phases 0–4) estimates 15–21 weeks with single-developer pacing. This accelerated plan compresses to **5–6 weeks** by combining:

1. **Expert team** that eliminates learning-curve delays on both the patterns and the codebase
2. **AI agents** that eliminate mechanical work (test generation, RM migrations, aggregate scaffolding)
3. **Parallel execution** across 3 senior tracks + 3 support tracks running simultaneously

The critical path is not agent throughput — it's the sequence of **human design decisions** that gate dependent work. The team composition eliminates most of those gates.

Team

Role	Strengths	Primary Responsibility
James (ES Expert A)	World-class event sourcing + large system design	Architect track — aggregate design, ProcessManager state machines, SFM extraction strategy
Chris (ES Expert B)	World-class event sourcing + large system design	Infrastructure track — ICheckpointStore, CachingRepository, ReadModelBase integration, PostgreSQL topology
Josh (Principal Engineer)	Knows the PowerModels codebase intimately	Integration + review — reviews all agent output, handles pipeline migration, catches domain edge cases
Farouk (Support 1)	Development	Test track — RestoreFromEvents tests, aggregate test suites
Tom (Support 2)	Development	Migration track — RM checkpoint migrations, SpreadsheetAdapter audit
Maeve (Support 3)	QA	Verification track — integration tests, regression, performance measurement

Agent Role

AI agents act as **force multipliers for the support team** and **scaffolding generators for the senior engineers**:

Work Type	Agent Contribution	Human Contribution
RestoreFromEvents tests	Generate complete test from Apply method signatures	Support 1 validates + runs
RM checkpoint migrations	Generate before→after code for each of 46 RMs	Support 2 validates build + behavior
Source constructor fixes	Generate `:base(source)` additions	Principal batch-reviews

New aggregate scaffolding	Generate aggregate + commands + events + handler	ES Expert reviews design, adjusts invariants
ProcessManager state machines	Generate skeleton from state enum	ES Expert fills in transition logic
Entry patterns / classification rules	Generate from domain spec	Support validates accounting correctness
Vault rescan + doc regeneration	Fully automated via VaultTool	None required

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Why This Team Compresses the Critical Path

Design decisions that normally gate weeks → resolved in hours

Decision	Normal Delay	This Team
Journal aggregate design (balanced entries, period validation)	1–2 weeks research + iteration	ES Expert A designs in hours — has built event-sourced ledgers at scale
ICheckpointStore integration into ReadModelBase	1 week (learn the infra, test the pattern)	ES Expert B likely wrote the infrastructure — wiring their own code
SFM extraction dual-write strategy	1–2 weeks (understand all 9 RM subscriptions)	Principal knows every RM subscription and handler route by heart
Reconciliation pipeline → ProcessManager migration	2 weeks (map 12 implicit steps)	Principal has lived with those 19 defects — the PM state machine formalizes what's already in their head
PostgreSQL topology for checkpoint + RM persistence	3–5 days ops decision	ES Expert B makes the call; `InMemoryCheckpointStore` is fallback

Mechanical work that normally fills weeks → agents batch in hours

Work	Manual Estimate	Agent-Assisted
27 RestoreFromEvents tests	2–3 weeks	2–3 days (agents generate, Support 1 validates)
46 RM checkpoint migrations	2–3 weeks	3–4 days (agents apply template, Support 2 validates)
4 source constructor fixes	1 day	1 hour
AccountingSystem extension (4 new commands)	3–5 days	1–2 days (agent scaffolds, ES Expert B reviews)
CoA template engine	1 week	2–3 days (agent generates, Support 1 validates)

Entry Patterns Library	1 week	2–3 days (agent generates patterns from accounting spec)
Classification rules enhancement	1 week	2–3 days

Week-by-Week Execution

Week 1: Safety Net + Infrastructure Start

```
ES Expert A     Phase 0.2 — SFM RestoreFromEvents test (87 Apply methods)
                ↓ if Apply bugs found → event versioning decisions (has EventVersioning infra)

ES Expert B     Phase 1.1 — ICheckpointStore integration into ReadModelBase
                (their own infrastructure — pattern validated by end of week)

Principal       Phase 0.1 — fix 4 source constructors (trivial)
                → reviews all agent-generated RestoreFromEvents tests
                → starts Phase 1.4 (wire ProcessManagerBase into solution)

Support 1       Phase 0.3 — RestoreFromEvents for 5 growth aggregates
                (agents generate each test, Support 1 validates + runs)
                Aggregates: ChartOfAccounts, AccountBalance, AccountingSystem,
                EntrySet, TasklistItem

Support 2       CI integration for new tests; build verification

Support 3       Baseline measurement: startup time at 1K, 5K, 15K events
                (before any changes — establishes the "before" number)
```

Week 1 Exit:

- 6/27 RestoreFromEvents tests passing (SFM + 5 growth targets)
- 4 source constructors fixed
- ICheckpointStore integration pattern validated
- ProcessManagerBase wired into solution
- Startup time baseline measured

Week 2: Infrastructure Complete + Domain Start

```
ES Expert A     Phase 2.2 — Journal aggregate design + implementation
                (starts immediately — doesn't wait for Phase 1 to fully complete)
                Commands: PostEntry, ReverseEntry, CorrectEntry
                Design constraints: bus-only coordination, no cross-aggregate GetById
                Ledger RM placement: ModelServer, not SpreadsheetAdapter

ES Expert B     Phase 1.2 — checkpoint 9 SFM RMs (agents generate migrations)
                Phase 1.3 — two-tier IReadModelStore (InMemory + Dapper)
                Phase 1.5 — CachingRepository for SFM GetById

Principal       Phase 1.4 complete (ProcessManagerBase available)
                → reviews 9 RM checkpoint migrations
```

```
                → starts Phase 2.1 (AccountingSystem extension — agent scaffolds)

Support 1       Phase 1.2 support — validates each of 9 RM migrations as agents produce them

Support 2       Phase 2.5 — SpreadsheetAdapter RM audit (agent-assisted catalog of 40 RMs)

Support 3       Post-Phase-1.2 measurement: SFM stream startup time with checkpoints
                Target: >80% reduction in handler invocations
```

Week 2 Exit:

- All 9 SFM RMs checkpointed (hottest stream mitigated)
- Two-tier RM persistence operational
- CachingRepository wrapping SFM
- Journal aggregate designed and under implementation
- AccountingSystem extension in progress
- SpreadsheetAdapter RM audit complete
- Startup time measured with checkpoints (before/after comparison)

Week 3: Domain Foundation Complete

```
ES Expert A     Phase 2.2 complete (Journal aggregate + tests)
                → Phase 2.4 — Reconciliation ProcessManager design
                  ReconciliationProcess : ProcessAggregateBase<ReconState>
                  8 states, timeout handling, compensation logic

ES Expert B     Phase 2.1 complete (AccountingSystem: entity type, industry, fiscal year)
                → Phase 2.3 (CoA template engine — agent scaffolds, expert reviews)
                → starts Phase 3.5 planning (SFM extraction dual-write strategy)

Principal       Phase 2.4b — reconciliation pipeline migration plan
                Maps each of 12 Pipeline.cs steps to PM state transitions
                Identifies which SpreadsheetAdapter RMs need re-subscription
                Feature flag design for gradual cutover

Support 1       Phase 2.3 support (CoA templates — agent generates, support validates)
                → starts Phase 3.6 (remaining RestoreFromEvents — agents batch-generate)

Support 2       Continue RM checkpoint batch (remaining 37 ReadModelBase RMs)
                Phase 4.1 pulled forward — agents apply same template as SFM RMs

Support 3       Journal aggregate integration tests
                AccountingSystem extension integration tests
                Reconciliation PM test harness setup
```

Week 3 Exit:

- Journal aggregate complete with RestoreFromEvents + full test suite
- AccountingSystem extended (5+ events)
- CoA template seeding working
- Reconciliation ProcessManager designed with full state machine
- Pipeline migration plan documented (12 steps → 8 PM states)

- ~20/46 RMs checkpointed (SFM 9 + batch progress)

Week 4: Process Managers + Feature Buildout Start

```
ES Expert A     Phase 2.4 — ReconciliationProcess implementation
                State machine, timeout handling, compensation
                + ReconciliationProcessManager (LoadOrCreate, SaveAndTakeCommands)
                CategoryStreamWarmup for $ce-reconProcessAgg

ES Expert B     Phase 3.5 — SFM table-mapping extraction begins
                Dual-write: events on both old (SFM) and new (dedicated) streams
                20 events moving to DataTableMap, ListDataTableMap, ManualTableMap

Principal       Phase 2.4b — pipeline migration execution
                Feature flag on; step-by-step wiring through PM
                Old pipeline as fallback until verified

Support 1       Phase 3.2 (Entry Patterns Library — agent generates patterns)
                Phase 3.3 (Classification Rules — agent scaffolds per-entity config)

Support 2       Continue RM checkpoint batch (~35/46 done)
                Phase 4.3 (ModelTemplateRm audit — does it need 82 events?)

Support 3       Reconciliation PM integration tests
                Pipeline migration regression tests (old vs new path comparison)
                Defect rate tracking setup
```

Week 4 Exit:

- Reconciliation ProcessManager operational (feature flag on, old pipeline as fallback)
- SFM extraction in progress (dual-write active)
- Entry Patterns Library operational
- Classification rules configurable per entity with 85/15 threshold
- ~35/46 RMs checkpointed

Week 5: Feature Completion + Extraction

```
ES Expert A     Phase 3.1 — PeriodClose ProcessManager
                PeriodCloseProcess : ProcessAggregateBase<CloseState>
                5 states, SoftClose/HardClose/Reopen, checklist coordination
                + ReconciliationState aggregate

ES Expert B     Phase 3.5 complete — SFM extraction verification
                All 9 RMs re-subscribed to correct streams
                SFM reduced from 31 → ~11 registered events
                → Phase 4.4 (FinancialModelService split: ~100 core + ~60 mapping)

Principal       Phase 3.4 — GL Report Generation (Trial Balance, P&L, Balance Sheet)
                Read model projections from Journal events
                All with ICheckpointStore
                → verifies reconciliation pipeline migration (flag fully on)
```

```
Support 1       Phase 3.6 — remaining RestoreFromEvents tests
                Agents generate all 21 remaining; Support 1 batch-validates
                Target: 27/27 + 3 new aggregates

Support 2       RM checkpoint batch complete (46/46)
                Helps with GL report RM implementation

Support 3       Full regression suite
                Reconciliation defect rate measurement (before/after)
                Startup benchmarks across all business sizes
                SFM extraction verification tests
```

Week 5 Exit:

- PeriodClose ProcessManager operational
- ReconciliationState aggregate created
- GL reports rendering from Journal events
- SFM extraction complete (31 → ~11 events, 87 → ~67 Apply methods)
- FinancialModelService split in progress
- 27/27 RestoreFromEvents tests + new aggregates
- 46/46 RMs checkpointed
- Reconciliation pipeline fully on ProcessManager (old path decommissioned)

Week 6: Consolidation + Sign-Off

```
ES Expert A     Architectural review of all Phase 2-3 output
                Verify: no new implicit sagas, all coordination via bus commands
                Sign-off on ProcessManager patterns

ES Expert B     FinancialModelService split complete
                Verify: all checkpoint + persistence coverage
                PostgreSQL operational review

Principal       Final integration review
                Verify: no regressions, feature flag cleanup
                Pipeline migration: old path decommission confirmed

Support 1       Final RestoreFromEvents verification (all pass)
                New aggregate test compliance check

Support 2       Documentation updates
                Vault rescan (VaultTool project-refresh)

Support 3       Performance baselining (Phase 4.5):
                - Startup time at 1K, 5K, 15K, 50K events
                - GetById latency for SFM at various stream sizes
                - Reconciliation defect rate comparison
                - RM replay invocation count (before/after)
```

Week 6 Exit = Plan Complete: All Phase 0–4 deliverables met.

Gantt Summary

```
         Week 1        Week 2        Week 3        Week 4        Week 5        Week 6
Expert A ██ P0.2 SFM ████ P2.2    ███ P2.4      ████ P2.4     ███ P3.1      █ Review
         RestoreFrom Journal Agg  Recon PM      Recon PM      PeriodClose
         Events      Design+Impl  Design        Implementation PM + ReconState

Expert B ████ P1.1   ████ P1.2/3/5 ██ P2.1+2.3  ████ P3.5     ███ P3.5      █ P4.4
         Checkpoint  9 RM migrate  AcctSys+CoA  SFM Extract   SFM Complete  FMS Split
         Store Integ Two-tier+Cache              Dual-write    Verify

Princpl  █ P0.1      ██ P1.4      ███ P2.4b     ████ P2.4b    ███ P3.4      █ Final
         Source ctors PM wiring    Pipeline      Pipeline      GL Reports    Review
         + reviews   + reviews    migration plan migration exec

Supp 1   ████ P0.3   ██ P1.2 help ██ P2.3+P3.6  ██ P3.2+3.3   ████ P3.6     █ Verify
         RestoreFrom RM migration CoA+tests     Entry+Rules   Batch tests
         6 aggregates validation  batch start

Supp 2   █ CI setup  ██ P2.5      ████ P4.1     ████ P4.1+4.3 ██ P4.1       █ Docs
                     SA RM audit  RM checkpoint  RM batch+     RM complete   Vault
                                  batch          TmplRm audit  46/46         rescan

Supp 3   ██ Baseline ██ Measure   ██ Integ      ████ Recon    ████ Full     ██ Final
         startup time checkpoint  tests          PM tests      regression    baselines
                     impact                      pipeline regr  + benchmarks
```

Deliverable Tracking

Phase 0 Deliverables (Week 1)

#	Deliverable	Owner	Agent Role	Exit Criteria
0.1	4 source constructors fixed	Principal	Generate fixes	CI green
0.2	SFM RestoreFromEvents test	ES Expert A	Scaffold from Apply methods	Test passes; any Apply bugs documented
0.3	5 growth aggregate RestoreFromEvents	Support 1	Generate complete tests	All 5 pass
—	Startup time baseline	Support 3	—	Measured at 1K, 5K, 15K events

Phase 1 Deliverables (Weeks 1–2)

#	Deliverable	Owner	Agent Role	Exit Criteria
1.1	ICheckpointStore in ReadModelBase	ES Expert B	Generate migration template	Optional param, null default, opt-in

1.2	9 SFM RMs checkpointed	ES Expert B + Support 1	Generate 9 migrations	>80% startup invocation reduction
1.3	Two-tier IReadModelStore	ES Expert B	Generate Dapper store config	InMemory (hot) + PostgreSQL (persisted)
1.4	ProcessManagerBase in solution	Principal	Namespace adaptation	Available as base class
1.5	CachingRepository for SFM	ES Expert B	—	GetById cached within session

Phase 2 Deliverables (Weeks 2–4)

#	Deliverable	Owner	Agent Role	Exit Criteria
2.1	AccountingSystem extended	ES Expert B	Scaffold commands + events	Entity type, industry, fiscal year, close policy
2.2	Journal aggregate	ES Expert A	Scaffold aggregate	PostEntry, ReverseEntry, CorrectEntry + RestoreFromEvents
2.3	CoA template engine	Support 1	Generate template + seeding	≥2 industry templates working
2.4	Reconciliation ProcessManager	ES Expert A	Scaffold state machine	8 states, timeouts, compensation, RestoreFromEvents
2.4b	Pipeline migration	Principal	—	Feature flag on, old pipeline as fallback
2.5	SpreadsheetAdapter RM audit	Support 2	Catalog 40 RMs	Consolidation candidates identified

Phase 3 Deliverables (Weeks 4–5)

#	Deliverable	Owner	Agent Role	Exit Criteria
3.1	PeriodClose PM + ReconciliationState	ES Expert A	Scaffold aggregates	5 states, SoftClose/HardClose/Reopen
3.2	Entry Patterns Library	Support 1	Generate patterns	≥6 accounting scenarios
3.3	Classification rules enhancement	Support 1	Scaffold config model	Per-entity rules, 85/15 threshold
3.4	GL Report Generation	Principal	Generate RM projections	Trial balance, P&L, balance sheet
3.5	SFM table-mapping extraction	ES Expert B	—	31→~~11 events, 87→~~67 Apply, dual-write verified

3.6	Remaining RestoreFromEvents (21)	Support 1	Generate all 21 tests	27/27 passing

Phase 4 Deliverables (Weeks 3–6, pulled forward)

#	Deliverable	Owner	Agent Role	Exit Criteria
4.1	All 46 RMs checkpointed	Support 2	Generate 37 migrations	46/46 checkpoint-based
4.3	ModelTemplateRm audit	Support 2	Analyze 82 subscriptions	Recommendations documented
4.4	FinancialModelService split	ES Expert B	—	Core (~~100 cmds) + TableMapping (~~60 cmds)
4.5	Performance baselines	Support 3	—	Startup <2s at 15K events; GetById <100ms at 5K

Success Metrics (Week 6 Targets)

Metric	Before	Target	Measured By
Calendar time to complete	—	≤6 weeks	Calendar
Startup time (15K events)	Baseline (Week 1)	<2s	Support 3 benchmarks
SFM GetById replay	Full stream (3,557+ events)	<100 events (snapshot)	CachingRepository metrics
RestoreFromEvents coverage	0/27	30/30 (27 existing + 3 new)	test-coverage.md
Source constructor coverage	23/27	30/30	aggregates.md
Explicit process managers	0	2 (Reconciliation, PeriodClose)	saga-catalog.md
RM checkpoint coverage	0/46	46/46	read-models.md
SFM registered events	31	~11	god-aggregate.md
SFM Apply methods	87	~67	scan-data.json
SFM lines	3,557	~1,500-2,000	scan-data.json
FinancialModelService commands	166	~80-90	message-map.md
Reconciliation defect rate	29%	<10%	defect-analysis.md
SpreadsheetAdapter defect rate	15%	<10%	defect-analysis.md

Implicit sagas	7	≤4	saga-catalog.md
Accountant workflow epics	0/5	5/5	GitHub issues
Missing aggregates	3	0	scan-data.json

Risk Factors That Could Extend to 7–8 Weeks

Risk	Likelihood	Impact	Trigger	Mitigation
SFM RestoreFromEvents reveals >5 Apply bugs	HIGH	+1 week	Week 1 test results	EventVersioning + VersionedEventSerializer ready. ES Expert A handles upcaster design.
PostgreSQL ops setup delays	MEDIUM	+3–5 days	Week 2 checkpoint deployment	`InMemoryCheckpointStore` as fallback; PostgreSQL deferred to Week 3.
Journal aggregate design iteration	LOW (expert team)	+1 week	Week 2 design review	ES Expert A has built event-sourced ledgers before. Start minimal (debit/credit/amount/date), extend via Entry Patterns.
Reconciliation pipeline migration regression	MEDIUM	+3–5 days	Week 4 feature flag activation	Old pipeline as fallback. Support 3 runs regression suite before flag-on.
SFM extraction breaks RM subscriptions	MEDIUM	+3–5 days	Week 4–5 dual-write phase	Principal knows all 9 RM subscriptions. Migrate one RM at a time. Full RestoreFromEvents re-verification.

The Acceleration Formula

```
Traditional (1 developer):     15–21 weeks
  → Limited by: sequential phases, learning curves, mechanical work

Agents only (no team context):  7–10 weeks
  → Limited by: human review cycles, design decisions waiting for expertise

This team + agents:             5–6 weeks
  → Limited by: sequential dependencies only (Journal before GL reports,
    CheckpointStore before RM migrations, ProcessManagerBase before PM implementation)
```

What the agents eliminate: All mechanical work — test generation, RM migrations, aggregate scaffolding, source constructor fixes, documentation, vault rescans.

What the experts eliminate: Design decision delays — Journal aggregate design, ProcessManager state machines, SFM extraction strategy, PostgreSQL topology, pipeline migration mapping.

What the principal eliminates: Review bottlenecks — knows every file, every RM subscription, every handler route. Agent output reviewed in minutes, not hours.

What remains on the critical path: The irreducible sequence of dependencies where each phase's output gates the next phase's input. With this team, those gates open in days, not weeks.